QUAID-I-AZAM UNIVERSITY ISLAMABAD

DEPARTMENT OF COMPUTER SCIENCE



DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

SOFTWARE PROJECT TECHNICAL DOCUMENTATION

NAME	FAIZAN AHMAD
REG#	04072213038

NAME	Zohaib Zafar
REG#	04072312049

Submitted to:

Dr. Umer Rashid

Abstract

Detection of clickbait through Machine Learning

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

Table Contents

Chapter No. 01 ..... **Error! Bookmark not defined.**

Software Project Management Plan ..... **Error! Bookmark not defined.**

1 Introduction ..... **Error! Bookmark not defined.**

1.1 Project Overview ..... **Error! Bookmark not defined.**

1.2 Project Deliverables ..... **Error! Bookmark not defined.**

2 Project Organization ..... **Error! Bookmark not defined.**

2.1 Software Process Model ..... **Error! Bookmark not defined.**

2.2 Roles and Responsibilities ..... **Error! Bookmark not defined.**

2.3 Tools and Techniques ..... **Error! Bookmark not defined.**

3 Project Management Plan ..... **Error! Bookmark not defined.**

3.1 Tasks ..... **Error! Bookmark not defined.**

3.1.1 Requirement Analysis Phase ..... **Error! Bookmark not defined.**

3.1.1.1 Requirements Identification ..... **Error! Bookmark not defined.**

3.1.1.2 Defining Use Cases ..... **Error! Bookmark not defined.**

3.1.1.3 Develop Domain Model ..... **Error! Bookmark not defined.**

3.1.1.4 ERD Diagram ..... **Error! Bookmark not defined.**

3.1.1.5 Software Requirements Specification Completion ..... **Error! Bookmark not defined.**

3.1.2 Design Phase ..... **Error! Bookmark not defined.**

3.1.2.1 Develop Design ..... **Error! Bookmark not defined.**

3.1.2.2 Developing Interfaces ..... **Error! Bookmark not defined.**

3.1.2.3 Class Diagram Designing ..... **Error! Bookmark not defined.**

3.1.2.4 Sequence Diagram Design ..... **Error! Bookmark not defined.**

3.1.2.5 Design Phase Verification ..... **Error! Bookmark not defined.**

3.1.2.6 Software Test Documentation ..... **Error! Bookmark not defined.**

3.1.3 Implementation Phase ..... **Error! Bookmark not defined.**

3.1.3.1 System Implementation ..... **Error! Bookmark not defined.**

3.2 Assignments ..... **Error! Bookmark not defined.**

3.3 Project Schedule ..... **Error! Bookmark not defined.**

3.4 Gantt Chart ..... **Error! Bookmark not defined.**

Chapter 2 ..... **Error! Bookmark not defined.**

Software Requirement Specification ..... **Error! Bookmark not defined.**

2.1 Introduction ..... **Error! Bookmark not defined.**

2.2 Functional Requirements ..... **Error! Bookmark not defined.**
Use Case Diagram ..... **Error! Bookmark not defined.**
***Use Case* Description** ..... **Error! Bookmark not defined.**
UC-1 User Registration ..... **Error! Bookmark not defined.**
UC-2 User Login ..... **Error! Bookmark not defined.**
UC-3 Reset Password ..... **Error! Bookmark not defined.**
UC-4 User Logout ..... **Error! Bookmark not defined.**
UC-5 Input Headlines / News Text ..... **Error! Bookmark not defined.**
UC-6 Upload Dataset ..... **Error! Bookmark not defined.**
UC-7 Train Machine Learning Model ..... **Error! Bookmark not defined.**
UC-8 Detect Clickbait ..... **Error! Bookmark not defined.**
UC-9 View Prediction Result ..... **Error! Bookmark not defined.**
UC-10 Explain Prediction ..... **Error! Bookmark not defined.**
UC-11 View Model Accuracy and Performance ..... **Error! Bookmark not defined.**
UC-12 Save Detection History ..... **Error! Bookmark not defined.**
UC-13 View Detection History ..... **Error! Bookmark not defined.**
System Sequence Diagram ..... **Error! Bookmark not defined.**
SSD UC-1 User Registration ..... **Error! Bookmark not defined.**
SSD UC-4 User Logout ..... **Error! Bookmark not defined.**
SSD UC-6 Upload Dataset ..... **Error! Bookmark not defined.**
SSD UC-7 Train Machine Learning Model ..... **Error! Bookmark not defined.**
SSD UC-8 Detect Clickbait ..... **Error! Bookmark not defined.**
SSD UC-9 Explain Prediction ..... **Error! Bookmark not defined.**
SSD UC-10 Save Detection History ..... **Error! Bookmark not defined.**
SSD UC-11 View Detection History ..... **Error! Bookmark not defined.**
Domain Model ..... **Error! Bookmark not defined.**
Chapter 3 ..... **Error! Bookmark not defined.**
Software Design and Description ..... **Error! Bookmark not defined.**
Sequence Diagrams ..... **Error! Bookmark not defined.**
SD UC-1 User Registration ..... **Error! Bookmark not defined.**
SD UC-2 User Login ..... **Error! Bookmark not defined.**
SD UC-3 Reset Password ..... **Error! Bookmark not defined.**
SD UC-4 Input Headline / News text ..... **Error! Bookmark not defined.**

SD UC-5 Text Preprocessing ..... **Error! Bookmark not defined.**
SD UC-7 Model Training ..... **Error! Bookmark not defined.**
SD UC-8 Clickbait Prediction ..... **Error! Bookmark not defined.**
SD UC-9 Explanation Generation ..... **Error! Bookmark not defined.**
SD UC-10 Save Detection History ..... **Error! Bookmark not defined.**
SD UC-11 View Detection History ..... **Error! Bookmark not defined.**
Activity Diagram ..... **Error! Bookmark not defined.**
Class Diagram ..... **Error! Bookmark not defined.**
Chapter No 4 6
Software Testing 6
Introduction ..... **Error! Bookmark not defined.**
Acceptance Test for UC-1 User Registration ..... **Error! Bookmark not defined.**
Acceptance Test for UC-2 User Login ..... **Error! Bookmark not defined.**
Acceptance Test for UC-3 Reset Password ..... **Error! Bookmark not defined.**
Acceptance Test for UC-4 User Logout ..... **Error! Bookmark not defined.**
Acceptance Test for UC-6 Upload Dataset ..... **Error! Bookmark not defined.**
Acceptance Test for UC-7 Train Machine Learning Model 7
Acceptance Test for UC-8 Detection Clickbait 7
Acceptance Test for UC-9 View Prediction Result 8
Acceptance Test for UC-10 Explain Prediction 9
Acceptance Test for UC-11 View Model Accuracy & Performance 10
Acceptance Test for UC-12 Save Detection History 11
Acceptance Test for UC-13 View Detection History 12
Chapter No 5 ..... **Error! Bookmark not defined.**
Conclusion and Future Enhancement ..... **Error! Bookmark not defined.**
Introduction ..... **Error! Bookmark not defined.**
Summary and Conclusion ..... **Error! Bookmark not defined.**
References ..... **Error! Bookmark not defined.**

Chapter No 4

Software Testing

Acceptance Test for UC-7 Train Machine Learning Model

Test ID	TC-7	
Description	Confirm that the gadget trains the machine studying version the use of the uploaded dataset and handles missing or invalid statistics as it should be.	
Setup	1. Ensure the Clickbait Detection System is going for walks. 2. Ensure the person is logged into the gadget. 3. Ensure a valid clickbait has already been uploaded to the device. 4. Ensure the Model Training alternative is out there from the dashboard or training interface.	
Instruction	Valid Input	1. Navigate to the Model Training segment. 2. Ensure a valid dataset is to be had inside the device. 3. Click the Train Model button. 4. Wait for the education method to complete.
	Invalid Input	1. Attempt to teach the model with out importing any dataset. 2. Attempt to train the version with an invalid or corrupted dataset. 3. Attempt to begin the schooling technique more than one instances concurrently.
Expected Result	Valid Input	• The device hundreds the uploaded dataset. • The dataset is processed and used to educate the gadget gaining knowledge of version. • The gadget completes the schooling manner correctly. • A confirmation message together with "Model schooling finished efficiently" is displayed.
	Invalid Input	1. If dataset isn't to be had: System presentations: "No dataset to be had for schooling". 2. If dataset is invalid: System presentations: "Dataset format isn't always legitimate for training". 3. If training is already in progress: System displays: "Model education is already strolling". 4. Training technique isn't always started.

Table 20

Acceptance Test for UC-8 Detection Clickbait

Test ID	TC-8
Description	Verify that the system analyzes the entered headlines or information text using the trained system learning version and efficaciously classifies it as

	Clickbait or Non- Clickbait, even as additionally managing empty or invalid enter nicely.	
Setup	1. Ensure the Clickbait Detection System is strolling. 2. Ensure the user is logged into the device. 3. Ensure the device gaining knowledge of model has already been skilled effectively. 4. Ensure the textual content input interface is available from the device dashboard.	
Instruction	Valid Input	1. Navigate to the Clickbait Detection section of the gadget. 2. Enter a headline in all likelihood to be clickbait, for instance: • "You Won't Believe What Happened Next" 3. Click the Analyze / Detect Clickbait button. 4. Enter a normal informational headline, for instance: • "Government Announces New Education Policy". 5. Click the Analyze / Detect Clickbait button once more.
	Invalid Input	1. Attempt detection without entering my text. 2. Enter most effective special characters or meaningless enter (e.G., @@@###). 3. Attempt detection when the gadget studying version is not to be had.
Expected Result	Valid Input	• The device procedures the enter text using herbal language processing techniques. • The skilled system learning model analyzes the headline. • The device presentations the class end result: o Clickbait, or o Non-Clickbait • The result is displayed certainly at the interface along side the prediction outcome.
	Invalid Input	1. If the enter subject is empty: System displays: "Please enter a headline or news text for evaluation". 2. If the input is invalid: System presentations the precise validation message. 3. If the version is unavailable: System presentations: "Prediction model is not to be had". 4. No prediction is generated invalid cases.

Table 21

Acceptance Test for UC-9 View Prediction Result

Test ID	TC-9

Description	Verify that the system displays the prediction end result certainly after analyzing the input headline or news text using the skilled machine studying model.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the user is logged into the system. 3. Ensure the device studying model has already been skilled. 4. Ensure a headline or information textual content has been submitted for clickbait detection.	
Instruction	Valid Input	1. Navigate to the Clickbait Detection interface. 2. Enter a legitimate headline or news text. 3. Click the Analyze / Detect Clickbait button. 4. Wait for the gadget to process the enter and generate the prediction.
	Invalid Input	1. Attempt to view the prediction result without performing clickbait detection. 2. Refresh the web page before the prediction method completes. 3. Attempt to view prediction results while the version has not been skilled.
Expected Result	Valid Input	• The system system the enter the use of the trained machine mastering model. • The gadget presentations the prediction end result on the interface. • The result without a doubt suggests whether or not the enter text is: o Clickbait, or o Non-Clickbait. • The prediction end result is displayed in a clear and readable layout for the person.
	Invalid Input	1. If detection has no longer been done: machine displays: “No prediction result to be had”. 2. If the prediction system is incomplete: System presentations a processing or ready message. 3. If the model is unavailable: System presentations: “Prediction version now not to be had”.

Table 22

Acceptance Test for UC-10 Explain Prediction

Test ID	TC-10
Description	Verify that the machine presents an cause of the prediction end result, highlighting the important thing words pr elements that stimulated the category.
Setup	1. Ensure the Clickbait Detection System is running. 2. Ensure the consumer is logged into the device. 3. Ensure the device getting to know version has been educated efficaciously.

	4. Ensure a headline or information textual content has already been analyzed and a prediction end result is to be had.	
Instruction	Valid Input	1. Navigate to the prediction end result page after appearing clickbait detection. 2. Click the explain prediction alternative. 3. Wait for the gadget to generate the reason of the prediction.
	Invalid Input	1. Attempt to view clarification with out appearing clickbait detection. 2. Attempt to get entry to explanation while no prediction end result exists. 3. Attempt to generate rationalization if the prediction technique has not finished.
Expected Result	Valid Input	• The gadget analyzes the prediction output. • The system highlights critical phrases or terms that stimulated the prediction. • The clarification is displayed really to the consumer to expose why the headline became classified as Clickbait or Non-Clickbait.
	Invalid Input	1. If prediction end result isn't always to be had: System displays: “No prediction to be had to give an explanation for”. 2. If explanation cannot be generated: System shows the correct errors message. 3. The system does no longer display rationalization without a valid prediction end result.

Table 23

Acceptance Test for UC-11 View Model Accuracy & Performance

Test ID	TC-11	
Description	Verify that the system efficiently shows the performance metrics of the educated machine mastering model, including accuracy and different assessment measures.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the consumer is logged into the machine. 3. Ensure the machine gaining knowledge of version has been successfully educated the usage of a legitimate dataset. 4. Ensure that the model assessment metrices had been generated after the education system.	
Instruction	Valid Input	1. Navigate to the version performance or version assessment segment of the system. 2. Select the choice view version accuracy and performance. 3. Observe the performance metrics displayed by the gadget.

	Invalid Input	1. Attempt to view model accuracy earlier than the model education process has been finished. 2. Attempt to access version overall performance metrics while no assessment information is available. 3. Refresh or reload the web page for the duration of the metrics loading manner.
Expected Result	Valid Input	• The system retrieves the version assessment metrics. • The system displays the overall performance signs sincerely on the interface. • The displayed metrics may also include: o Accuracy o Precision o Recall o F1-score • The metrics are offered in a readable and structured format for the person.
	Invalid Input	1. If the version has not been trained: device displays: “Model performance records isn't always available”. 2. If evaluation metrics are lacking: System displays the perfect notification message. 3. The machine does now not show wrong or empty overall performance statistics.

Table 24

Acceptance Test for UC-12 Save Detection History

Test ID	TC-12	
Description	Verify that the system shops clickbait detection outcomes inside the records, inclusive of the headline text, prediction end result, and detection timestamp.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the person is logged into the system. 3. Ensure the gadget learning model has been trained correctly. 4. Ensure the user has done clickbait detection on a headline or news text.	
Instruction	Valid Input	1. Navigate to the Clickbait detection interface. 2. Enter a valid headline or information textual content for evaluation. 3. Click the Analyze / Detect Clickbait button. 4. Wait for the prediction Result to appear. 5. Confirm that the gadget routinely saves the detection result in the history phase.
	Invalid Input	1. Attempt to shop detection history without appearing clickbait detection. 2. Attempt to store detection facts while the prediction procedure fails.

		3. Attempt to store empty or invalid detection data.
Expected Result	Valid Input	• The gadget successfully stores the detection bring about the records database. • The saved document Includes: o Input headline or new textual content. o Prediction end result (Clickbait / Non-Clickbait). o Detection date and time. • A confirmation message which includes "Detection end result saved efficiently" can be displayed.
	Invalid Input	1. If no detection has been finished: System does no longer create a history record. 2. If detection fails: System does not keep incomplete statistics. 3. The machine shows the ideal notification indicating that the file can not be saved.

Table 25

Acceptance Test for UC-13 View Detection History

Test ID	TC-13	
Description	Allow users to view formerly stored clickbait detection outcomes, displaying the headline, prediction end result, and detection time.	
Setup	1. Ensure the Clickbait Detection System is walking. 2. Ensure the person is logged into the device. 3. Ensure that at the least one clickbait detection end result has been stored inside the system records.	
Instruction	Valid Input	1. Navigate to the detection history section from the machine dashboard. 2. Select the option view detection history. 3. Observe the list of formerly stored detection information.
	Invalid Input	1. Attempt to view detection history whilst no detection information exists in the machine. 2. Attempt to access detection history without logging into the machine. 3. Refresh or reload the records web page for the duration of the loading manner.
Expected Result	Valid Input	• The machine retrieves the stored detection history from the database. • The device presentations the previous detection statistics in a established layout. • Each file have to consist of: o The analyzed headline or news textual content. o The prediction end result (Clickbait / Non-Clickbait).

		○ The date and time of detection.
	Invalid Input	1. If no records information exsist: System displays: “No detection history available”. 2. If the consumer isn't always logged in: System redirects the person to the login web page. 3. The gadget should now not displayed empty or incorrect history records.

Table 26